

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template - AT - 18 points

****The requirements from Element C will be used for Elements D, E, F, and H.****

Element C: Presentation and Justification of Solution Design Requirements

Durable Enough (HIGH Priority)

The design requirement for the prototype would have to resist water damage, expansion, and contraction due to temperature changes.

This design requirement directly ties to research from element B, as one of the patented solutions to road damage specifically focused on the structural stability of asphalt, which protects from environmental damage.

The design requirement was approved by the stakeholder, who is someone that is dependent on the quality of the roads in their daily life. They have an interest in maintaining that quality as they have a stake in it.
The stakeholder asked questions about how the mentioned durability and weather resistance can be achieved, and what exactly we would be filling the potholes with.

Design Requirement #2 Efficiency (HIGH Priority)

The asphalt would need to be a square yard road patch that should be run over for half an hour by a device that is 5% of the car's actual weight.

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This design is relevant to prior research because it is a road infrastructure as stated in element A and it is an asphalt road surface, not a tunnel or bridge structure as state in element B.

The design requirement was approved by the stakeholder who is a driver of suburban and rural roads and is annoyed by the potholes.

High Quality #3 High Quality (HIGH Priority)

The solution should be high quality in all aspects, such as making it durable, clean, effective, as well as balancing being easy & cheap to make, including having good material & longevity in use.

Measurement standards: The solution should last for at least 2 to 3 years without any changes to its structure, should have zero net impact on the environment, & should be reasonably cheap according to its material volume.

Most solutions had faults especially in quality, with different solutions having different problems due to not being durable, clean, or effective, as well as not being cheap or simple.

The design requirement was approved by the stakeholder.



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Aesthetically Appropriate #4 (LOW Priority)

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.

The design has to look aesthetically appropriate, like being the same color as the road instead of bright red.

This design requirement directly ties to prior research from Element B because it is clear that no cars would want to drive on a random patch of road that is a different color as they might mistake it for being broken. (Research from Element B showed us that we change one patch of a road)

The design requirement was approved by the stakeholder. They said they would also be concerned if they saw a random patch of color on the road and it might cause their attention to drift away from driving.

For design requirement 5, are you talking about a substance above the asphalt? What is asphalt made of?

Design Requirement #5 Grip (LOW Priority)

The texture of the asphalt used has to have enough friction/grip to ensure that tires don't



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slip.

This design requirement directly ties to prior research from Element A and discussions with a stakeholder because road friction is vital for tires to provide the grip necessary for safe vehicle control, allowing a car to accelerate, brake, and turn effectively. Without adequate friction, tires would slip and lose traction, leading to skidding, longer stopping distances, and an inability to control the vehicle's direction. My father, the stakeholder, described how hydroplaning and tire slipping off the road are one of the main reasons for accidents. They are connected to the problem because he drives every day from work to home, and roads are a daily part of his life. From research in Element A, we know that bikers also use the road. Without proper road friction, the bike's tires may also get very slippery, especially in wet conditions. Therefore, the design requirement directly ties to my prior research and my discussion with the stakeholder. The non-slip quality of asphalt comes from its surface texture and aggregate, which create friction. We need the asphalt that we create to contain more materials, such as calcined bauxite, ground tire rubber, or added aggregates.

The stakeholder approved the design requirement.

Rubric

Element C: Presentation and Justification of Solution Design Requirements

	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C1. Determination of design requirements. _____ design requirements are related to prior research.	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0



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Design requirements _____.	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized, but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or overly general	are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5 Five	score = 4 Four	score = 3 Three	score = 2 Two	score = 1 One	score = 0 None
C4. Objective & measurable. Design requirements _____.	score = 5 are consistently objective and measurable	score = 4 are nearly always objective and measurable	score = 3 are generally objective and measurable	score = 2 may be only sometimes objective and/or measurable	score = 1 are insufficiently measurable	score = 0 are not measurable
C5. Leading to solution. Design requirements _____.	score = 5 would be highly likely to lead to a tangible and viable problem solution	score = 4 would be likely to lead to a tangible and viable problem solution	score = 3 has the potential to lead to a tangible and viable problem solution	score = 2 has some potential to lead to a tangible and viable problem solution	score = 1 have little potential to lead to a tangible and viable problem solution	score = 0 cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 no primary stakeholder groups.

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stakeholder groups.						
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Soil material fit for reducing/ preventing road damage-- doesn't change shape when absorbing water, and resist temperature changes and weights. --

